FIT3080 Artificial Intelligence
Practice Questions for Final Exam – Machine Learning

1. **[Supervised Machine Learning]** Consider the following dataset consisting of 3 binary features and a binary class variable.

Feature 1	**Feature 2**	**Feature 3**	**Class**
1	0	0	0
1	0	1	1
1	1	1	1
1	1	1	1
0	1	1	0
0	0	1	0

   (a) Based on the dataset above, estimate the prior probability of Class 1.

   (b) What is the initial entropy of the class labels over all the data? Show the formula before plugging-in numbers.

   (c) If we were to split the data into 2 groups based on the value of Feature 1, what would be the entropy for each group?

   (d) What is the Information Gain of Feature 1?

   (e) You are told the Information Gain for Feature 2 and Feature 3 **after splitting on Feature1** is 0.08 and 0.19 respectively. Draw a decision tree for this problem.

   (f) We will now build a Naïve Bayes model for predicting the class from these data. Draw the Bayesian Network that corresponds to the Naïve Bayes model.

   (g) From the data table, write down the conditional probability estimates required for the Naïve Bayes model.

   (h) What class does the Naïve Bayes model predict for a new data item with values (1, 1, 1) for Features 1, 2 and 3 respectively? What is the probability of each class? How does the Naïve Bayes classification compare to the decision tree classification?

2. **[Decision Tree]** Assume that you are given the set of labeled *training examples* below, where each of three features has three possible values: a, b, or c. You choose to learn a decision tree from these data.

	F1	F2	F3	Output
ex1	c	b	b	+
ex2	a	a	c	+
ex3	b	c	c	+
ex4	b	c	a	−
ex5	a	b	c	−
ex6	c	a	b	−

   What score would the *information gain* calculation assign to feature F1, when deciding which feature to use as the root node for a decision tree being built?

3. [**Naïve Bayes**] Consider a traffic-monitoring agent trying decide whether the stoplights at an intersection are working or not ($W = w$ or $\neg w$). The agent observes two variables, the East-West light's color EW, which is either red or green, and the North-South light's color NS, which is also either red or green. When the lights are functioning, exactly one of the lights is green, while when the lights are broken, both are red (and flashing, but the agent cannot perceive this distinction). The agent has several observations as training data:

NS	EW	W
r	g	w
g	r	w
r	r	$\neg w$
g	r	w
r	g	w
r	g	w
g	r	w

   (a) Assume the agent uses a Naïve Bayes classifier to make its prediction. What are the parameters of the model?

   (b) What is the maximum likelihood estimate for the parameters?

   (c) Assume the agent observes that both lights are red. What is the probability that the lights are working according to the agent's model?

4. [**Perceptron**] Our aim is to train a perceptron with three features f_1, f_2, f_3 and three classes x, y, z. Suppose at some stage in the training, the weights for different classes are as follows:

$$\begin{aligned} w_x &= (-1, -1, -1) \\ w_y &= (1, 1, 1) \\ w_z &= (1, -1, 1) \end{aligned}$$

   Write down the updated weights after seeing a new training instance (1,1,1) with label z.

5. [**K-nearest-neighbour**] Assume we have the age, loan amount and pay status of some customer from the bank. We need to predict the pay status of a new loan applicant given only age and loan amount. The data are shown below:

Age	Loan Amount (K)	Pay Status
20	20	Paid
25	45	Paid
22	95	Default
35	53	Paid
35	120	Paid
33	150	Default
40	54	Default
45	75	Paid
52	20	Paid
49	220	Default
60	100	Default

A new bank customer with age 42 comes in and applies for a loan AUD$140K, how would the 3-nearest-neighbour algorithm predict the pay status of this customer? Paid or Default?

6. [**K-means clustering**] Consider the k-means clustering algorithm when answering the following questions.

   (a) What are the two steps performed at each iteration of the clustering algorithm?

   (b) Is the algorithm guaranteed to converge (stop) at some point? And if so, does it converge to a global minimum or a local minimum?

   (c) Choosing the "right number" of clusters for k-means clustering can be difficult. One strategy would be to choose the number of clusters that minimises the sum of the (squared) distances between data points and their cluster centres. Would this be a good strategy? Why?

   (d) Before running a clustering algorithm it is important to preprocess your data. What preprocessing is most important and why?

   (e) Assume we have following data with two attributes, apply the K-means clustering algorithm where K=2 and initialized cluster centroid C1=(1.25, 1.25) and C2=(1.75,4.75). You should use Euclidean distance to perform your calculations.

Instance	D1	D2
X1	1.0	1.5
X2	1.0	4.5
X3	2.0	1.5
X4	2.0	3.5
X5	3.0	2.5
X6	5.0	6.0
X7	4.0	5.5
X8	6.0	1.5